

Multivariable Calculus

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1 Preliminaries

By way of introducing the subject, we first introduce several pertinent operations that will recur throughout the subject. Accordingly, we first redefine the many concepts in a higher Euclidean space, such as

1.1 Dot Product

Definition 1.1.1. For vectors $\mathbf{a} = (a_1, a_2, \dots, a_n)$ and $\mathbf{b} = (b_1, b_2, \dots, b_n)$, their **dot product** is defined by $\mathbf{a} \cdot \mathbf{b} := \sum_{i=1}^n a_i b_i$. The dot product is a scalar.

Theorem 1.1.2. For a fixed vector $\mathbf{a} = (a_1, \dots, a_n)$, $\|\mathbf{a}\|^2 = \mathbf{a} \cdot \mathbf{a} = \sum_{i=1}^n a_i^2$.

Theorem 1.1.3. For two given vectors of the same dimension, $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{b} = (b_1, \dots, b_n)$, $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$.

Proof. By the Law of Cosines, if $\mathbf{c} = \mathbf{a} - \mathbf{b}$, then

$$\|\mathbf{c}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\| \|\mathbf{b}\| \cos \theta.$$

On the other hand,

$$\begin{aligned} \|\mathbf{c}\|^2 &= \mathbf{c} \cdot \mathbf{c} = (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) \\ &= \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2(\mathbf{a} \cdot \mathbf{b}). \end{aligned}$$

Comparing the two expressions yields

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta. \quad \square$$

Corollary 1.1.3.1. For two given vectors $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}$ is orthogonal to $\mathbf{b}$ iff $\mathbf{a} \cdot \mathbf{b} = 0$.

Example 1.1.4. (pset 1 partII p2) Let $\mathbf{u}, \mathbf{v}$ be vectors.

(a) Show that

$$\frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2) = \mathbf{u} \cdot \mathbf{v}.$$

(b) Given two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, find the unit vector which bisects the smaller angle between them.

Solution. (a) We proceed by expanding the left side: $\text{LHS} = \frac{1}{4}((\mathbf{u}^2 + 2\mathbf{u} \cdot \mathbf{v} + \mathbf{v}^2) - (\mathbf{u}^2 - 2\mathbf{u} \cdot \mathbf{v} + \mathbf{v}^2)) = \mathbf{u} \cdot \mathbf{v}$.

(b) Define $\hat{\mathbf{x}} := \frac{\mathbf{x}}{\|\mathbf{x}\|}$. The answer is

$$\frac{\hat{u} + \hat{v}}{\|\hat{u} + \hat{v}\|^2}$$

□

1.2 Dot Product Applications

1. One of the first fundamental applications of dot product is converting between lengths and angles and coordinates.

Example 1.2.1. Consider points $P, Q, R \in \mathbb{R}^3 : P = (1, 0, 0), R = (0, 0, 2), Q = (0, 1, 0)$. Find $\angle RPQ$.

Solution of Example 1.2.1. Express in terms of vectors, $\overrightarrow{PR} = \vec{R} - \vec{P} = (-1, 0, 2)$ and $\overrightarrow{PQ} = \vec{Q} - \vec{P} = (-1, 1, 0)$. Therefore, $\theta := \angle RPQ$, $\overrightarrow{PR} \cdot \overrightarrow{PQ} = \|\overrightarrow{PR}\| \|\overrightarrow{PQ}\| \cos \theta$. Thus,

$$\theta = \cos^{-1} \left(\frac{(-1, 0, 2) \cdot (-1, 1, 0)}{\|(-1, 0, 2)\| \|(-1, 1, 0)\|} \right) = \frac{1}{\sqrt{10}}. \quad \square$$

Example 1.2.2 (Pset 1 Part II p1). Find the dihedral angle between two faces of a regular tetrahedron.

Solution. Since all of the dihedral angles are equal (regular tetrahedron), we only need to find one such angle.

Method 1: Fix one face as the base and have its center to be the origin in the $\mathbb{R}^3$ space while it lies in the x-y plane with one vertex at $(0, 1, 0)$.

$$A := (0, 1, 0), B := \left(0, -\frac{1}{2}, 0\right), C := (0, 0, \sqrt{2}).$$

Then the desired angle $\angle ABC$ could be computed as Example 1.2.1.

Method 2: Notice the dihedral angle between two planes is equal to the angle between their normal vectors, we only need to compute the angle between two normal vectors.

$$\overrightarrow{BC} = (0, -\sqrt{3}, 0), \quad \overrightarrow{BD} = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}, \sqrt{2}\right),$$

$$\mathbf{n}_2 = \overrightarrow{BC} \times \overrightarrow{BD} = \left(-\sqrt{6}, 0, -\frac{\sqrt{3}}{2}\right) \sim (-2\sqrt{2}, 0, -1).$$

Thus, the angle could be computed with Theorem 1.1.3. □

2. Dot product is pivotal in dealing with surfaces.

Example 1.2.3. Fix $a, b, c \in \mathbb{R}$, show that the surface defined by $ax + by + cz = 0$ is a plane.

Solution of Example 1.2.3. Notice that the surface definition is

$$\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : \begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \right\}.$$

Geometrically, this is all vectors that are orthogonal to a fixed vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$, therefore a plane. $\square$

Corollary 1.2.3.1. For a plane defined by $ax + by + cz = 0$, it's normal vector is $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$.

This result could be generalized to surfaces beyond planes, See Theorem 2.6.2 and Theorem 2.2.2.

Theorem 1.2.4. Three planes intersect at a point iff their normal vectors are coplanar.

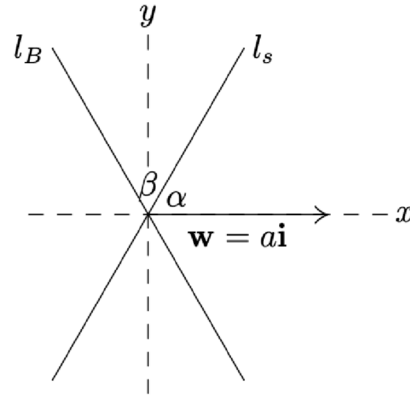
Proof. Left as an exercise. $\square$

3. In Physics, when dealing with force, dot product is often critical.

Example 1.2.5.

Background: By carefully positioning the sail the boat can be made to sail into the wind – this process is called *tacking*.

Described mathematically, the wind vector is first projected onto the perpendicular to the sail to obtain the direction of the force on the sail. This resultant force is then projected onto the axis of the boat to determine the direction in which the boat is being pushed. By orienting the sail correctly, this double projection can result in a vector with a component pointing into the wind.



In the figure, $w = ai$ is the wind direction. The line l_s is perpendicular to the sail ($0 \leq \alpha < \frac{\pi}{2}$), and the line l_B lies along the boat's axis ($0 \leq \beta < \frac{\pi}{2}$).

- a) Let $\mathbf{w}_1$ be the projection of $\mathbf{w}$ onto the line l_s . Show that $\mathbf{w}_1$ does not have a non-zero component in the direction opposite $\mathbf{w}$. (It is sufficient to show the projections on the sketch.)
- b) Find the projection of $\mathbf{w}_1$ onto l_B . (Give an explicit formula in terms of α and β .) What is the condition on α and β that this projection has a component in the $-\hat{i}$ direction? (For warmup you might try the specific case $\alpha = \frac{\pi}{3} = \beta$.)

Solution. (a) omitted due to simplicity. (b) The magnitude is

$$\|a\hat{i}\| \sin \alpha \sin \beta = a \sin \alpha \sin \beta,$$

while the direction is along l_B . geometrically, there is a $-\hat{i}$ component iff $\alpha + \beta > 90^\circ$ $\square$

1.3 Matrices

One of the motivations to have matrices is when transforming coordinate systems.

$$\begin{cases} u_1 = 2x_1 + 3x_2 + 3x_3 \\ u_2 = 2x_1 + 4x_2 + 5x_3 \\ u_3 = x_1 + x_2 + 2x_3 \end{cases}$$

which can be written as

$$\begin{pmatrix} 2 & 3 & 3 \\ 2 & 4 & 5 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$$

Definition 1.3.1. Let $\mathbf{A} = (a_{ij})$ be an $m \times n$ matrix and $\mathbf{B} = (b_{ij})$ be an $n \times p$ matrix.

The **matrix product** $\mathbf{AB}$ is the $m \times p$ matrix whose (i, j) -entry is

$$(\mathbf{AB})_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Equivalently, the (i, j) -entry of $\mathbf{AB}$ is obtained by taking the dot product of the i th row of $\mathbf{A}$ with the j th column of $\mathbf{B}$.

The product $\mathbf{AB}$ is defined only when the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$.

Definition 1.3.2. For a 2×2 matrix

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

its **determinant** is defined by

$$\det \mathbf{A} = ad - bc.$$

Definition 1.3.3. For a 3×3 matrix

$$\mathbf{A} = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix},$$

its **determinant** is defined by

$$\det \mathbf{A} = a_1 \det \begin{pmatrix} b_2 & b_3 \\ c_2 & c_3 \end{pmatrix} - a_2 \det \begin{pmatrix} b_1 & b_3 \\ c_1 & c_3 \end{pmatrix} + a_3 \det \begin{pmatrix} b_1 & b_2 \\ c_1 & c_2 \end{pmatrix}.$$

Definition 1.3.4. Let $\mathbf{A}$ be a square matrix. The (i, j) -**minor** of $\mathbf{A}$, denoted M_{ij} , is the determinant of the matrix obtained from $\mathbf{A}$ by deleting the i th row and the j th column.

Definition 1.3.5. Let $\mathbf{A}$ be a square matrix. The (i, j) -**cofactor** of $\mathbf{A}$, denoted C_{ij} , is defined by

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Definition 1.3.6. Let $\mathbf{A} = (a_{ij})$ be an $m \times n$ matrix. The **transpose** of $\mathbf{A}$, denoted $\mathbf{A}^T$, is the $n \times m$ matrix whose (i, j) -entry is

$$(\mathbf{A}^T)_{ij} = a_{ji}.$$

The determinant measures how a square matrix scales signed area in two dimensions and signed volume in three dimensions.

Matrix algebra satisfies many properties for a nice algebraic structure.

1. Associative

Theorem 1.3.7. In other words, if $\mathbf{X}$ is a vector and $\mathbf{A}$ and $\mathbf{B}$ are matrices,

$$(\mathbf{AB})\mathbf{X} = \mathbf{A}(\mathbf{BX}).$$

This theorem could be interpreted with the lens of matrices as transformations. The transformation $\mathbf{AB}$ is equivalent to applying the transformation described by $\mathbf{A}$ first, then transformation described by matrix $\mathbf{B}$.

2. Commutative However, note that $AB \neq BA$, which makes sense after the above theorem. In fact, given AB , BA is not even necessarily well-defined.
3. Multiplicative Identity There exist an unique identity fixing the height of the matrix being multiplied, with 1 along the diagonal and 0 elsewhere. Denote it as $\mathbf{I}$, we have

$$\mathbf{I}\mathbf{X} = \mathbf{X}.$$

1. Multiplicative Inverse

Definition 1.3.8. For a matrix A , matrix M is its inverse iff $AM = MA = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix.

Observe that a matrix has an inverse only if it is a square matrix, meaning it has the same number of rows and columns. In fact, any square matrix with a non-zero determinant has an inverse. Furthermore, finding the solution to a system of linear equations is equivalent to finding the inverse of a matrix. Consider a system of linear equations regarding n variables. If A is an n -by- n square matrix representing the coefficients of the variables in each equation, $\mathbf{X}$ a vector representing the solutions, and $\mathbf{B}$ representing the vector of the constants on the right side of the equations, then we have

$$A\mathbf{X} = \mathbf{B} \implies A^{-1}A\mathbf{X} = A^{-1}\mathbf{B} \implies \mathbf{X} = A^{-1}\mathbf{B}$$

Thus, how to find the inverse of a matrix becomes a natural question. Here, I claim, without proof, that if the adjunct matrix of a square matrix A is obtained by taking the transpose of the co-factor of the minors of A , then

$$A^{-1} = \frac{1}{\det A} \text{adj}(A).$$

Example 1.3.9. Consider the matrix

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$R \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix},$$

so R rotates every vector in the plane by 90° counterclockwise about the origin.

In particular,

$$R\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and

$$R\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

Thus, a system described by $\hat{i}$ and $\hat{j}$ is mapped to a system of $\hat{j}$ and $-\hat{j}$. Therefore, 90° counterclockwise.

1.4 Cross Product

With Definition 1.1.1, we have the means to often conveniently express angles and lengths in terms of vectors. However, there is another important concept that we are unable to grapple with using dot product. And that is area. For which we seek help from the use of (1.4.4).

Theorem 1.4.1. *The area of the triangle formed by two vectors with the same dimensions, $\mathbf{a}$ and $\mathbf{b}$ is $\det\begin{pmatrix} a_1 & a_2 \\ b_1 & b_2 \end{pmatrix}$*

Proof. Let θ denote the angle between $\mathbf{a}$ and $\mathbf{b}$. Then geometrically, we have the area to be

$$\frac{1}{2}\|\mathbf{a}\|\|\mathbf{b}\|\sin\theta.$$

Therefore, we aim to find $\sin\theta$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

Let $\mathbf{a}'$ be the image of rotation of $\mathbf{a}$ for $\frac{\pi}{2}$ in the direction towards $\mathbf{b}$. Let θ' denote the angle between $\mathbf{a}$ and $\mathbf{b}$.

We have

$$\|\mathbf{a}\|\|\mathbf{b}\|\sin\theta = \|\mathbf{a}'\|\|\mathbf{b}\|\cos(\theta') = \mathbf{a}' \cdot \mathbf{b}.$$

Let $\mathbf{a}$ have components (a_1, a_2) , then $\mathbf{a}'$ have components $(-a_2, a_1)$. Thus, the area is

$$\frac{1}{2}\mathbf{a}' \cdot \mathbf{b} = \frac{1}{2}(-a_2b_1 + a_1b_2) = \frac{1}{2}\det\begin{pmatrix} a_1 & a_2 \\ b_1 & b_2 \end{pmatrix}.$$

□

Remark. However, in this way, the area is not necessarily positive. Thus, we adopt the convention of signed area: the area formed by $\mathbf{a}$ rotating counterclockwise to $\mathbf{b}$ is positive by convention.

By the determinant definitions from the matrices section, the three-dimensional determinant gives the signed volume of the parallelepiped formed by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.

Now, we are ready to define cross product.

Definition 1.4.2. For two given three dimensional vectors $\mathbf{a}$ and $\mathbf{b}$ with components (a_1, a_2, a_3) and (b_1, b_2, b_3) , respectively. We define the cross product between $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a} \times \mathbf{b}$, as the vector

$$\det \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix}. \quad (1.4.4)$$

Theorem 1.4.3. For two given three dimensional vectors $\mathbf{a}$ and $\mathbf{b}$, $\|\mathbf{a} \times \mathbf{b}\|$ is equal to the (signed) area of the parallelogram formed by $\mathbf{a}$ and $\mathbf{b}$. And the direction of $\mathbf{a} \times \mathbf{b}$ is orthogonal to the plane formed by $\mathbf{a}$ and $\mathbf{b}$, while satisfying the right hand rule.

Proof. This result comes directly from the definition of the cross product. $\square$

Theorem 1.4.4. For three 3D vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$,

$$\begin{aligned} \det(\mathbf{a}, \mathbf{b}, \mathbf{c}) &= \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \\ &= \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) \\ &= \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}). \end{aligned}$$

Proof. This result comes directly from the expansion of both sides. However, an argument based on the geometric meaning in terms of the area of the parallelepiped could also be established. $\square$

Example 1.4.5 (Pset2 PartII P1). Suppose we know that when the three planes P_1 , P_2 , and P_3 in $\mathbb{R}^3$ intersect in pairs, we get three lines L_1 , L_2 , and L_3 which are *distinct* and *parallel*.

- Sketch a picture of this situation.
- Show that the three normals to P_1 , P_2 , and P_3 all lie in one plane, using a geometric argument.
- Show that the three normals to P_1 , P_2 , and P_3 all lie in one plane, using an algebraic argument. (Note that the three planes clearly do *not* all intersect at one point.)

Solution. (a) Graph omitted, the intersection of the three planes form a triangular prism. (c) Let the normals of the three planes be $\mathbf{n}_1$, $\mathbf{n}_2$, and $\mathbf{n}_3$, respectively. The pairwise intersections of the three planes have direction

given by $\mathbf{n}_1 \times \mathbf{n}_2$, $\mathbf{n}_2 \times \mathbf{n}_3$, and $\mathbf{n}_3 \times \mathbf{n}_1$. The three lines are parallel gives all of these quantities are equal.

$$\mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{n}_2 \times \mathbf{n}_3 \implies -\mathbf{n}_2 \times \mathbf{n}_1 = \mathbf{n}_2 \times \mathbf{n}_3 \implies -\mathbf{n}_1 \sim \mathbf{n}_3 \implies \mathbf{n}_1 \sim \mathbf{n}_3.$$

Similarly, $\mathbf{n}_2 \sim \mathbf{n}_1$, so all three normals are coplanar. $\square$

Example 1.4.6 (Pset2 PartII P3). For any plane P which is not parallel to the x - y plane, define the *steepest direction* on P to be the direction of any vector which lies in P and which makes the largest (acute) angle with the x - y plane.

a) Let P be the plane through the origin with normal vector $\mathbf{n}$. Derive a formula, in terms of $\mathbf{n}$, for a vector $\mathbf{w}$ which points in the steepest direction on P .

b) Now let P be the plane through the origin which contains two non-parallel vectors $\mathbf{u}$ and $\mathbf{v}$, where $\mathbf{u}$ and $\mathbf{v}$ do not both lie in the x - y plane. Derive a formula, in terms of $\mathbf{u}$ and $\mathbf{v}$, for a vector $\mathbf{w}$ which points in the steepest direction on P .

Solution. (a) We may draw some insights from previous sections. From Example 1.2.3, we have that $\mathcal{P}$ is described by

$$(a \ b \ c) \cdot \hat{\mathbf{n}} = 0.$$

Also, the desired direction is represented by ∇F , where $F(x, y, z) = 0$ describes the surface.

However, to fully solve the problem with cross product, we consider the two planes in play: $\mathcal{P}$ and the x - y plane. The desired vector, $\mathbf{v}$ satisfies the constraint of $\mathbf{v} \cdot \hat{\mathbf{n}} = 0$ and optimizes for the greatest value of $\mathbf{v} \cdot (\hat{\mathbf{i}} \times \hat{\mathbf{j}})$.

Let the normal vector of the x - y plane be $\mathbf{k}$, then $\mathbf{v} = (\mathbf{n} \times \mathbf{k}) \times \mathbf{n}$

(b) Notice that the $\hat{\mathbf{n}} \sim \mathbf{u} \times \mathbf{v}$, so we can directly apply the result from part (a) that

$$\mathbf{v} = ((\mathbf{u} \times \mathbf{v}) \times \mathbf{k}) \times (\mathbf{u} \times \mathbf{v}).$$

$\square$

Example 1.4.7. Fix three 3D dimensional vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$. Determine whether another vector $\mathbf{v}$ is on the plane formed by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.

Solution. The vector $\mathbf{v}$ is on the plane iff

$$\mathbf{v} \cdot ((\mathbf{a} - \mathbf{b}) \times (\mathbf{a} - \mathbf{c})) = 0.$$

Symmetrically, it could be expressed as

$$\det(\mathbf{p} - \mathbf{a} \quad \mathbf{p} - \mathbf{b} \quad \mathbf{p} - \mathbf{c}) = 0.$$

□

1.5 Parametric Equations

Previously, we described lines in $\mathbb{R}^3$ as intersections of planes. However, we may also describe a line as the trajectory of a moving point.

Suppose a line passes through the points

$$Q_0 = (-1, 2, 2)$$

and

$$Q_1 = (1, 3, -1).$$

Let $Q(t)$ denote a point moving along the line such that

$$Q(0) = Q_0$$

and

$$Q(1) = Q_1.$$

Then

$$(Q(t) - Q_0) = t(Q_1 - Q_0).$$

Since

$$(Q_1 - Q_0) = (1, 3, -1) - (-1, 2, 2) = (2, 1, -3),$$

we obtain

$$Q(t) = Q_0 + t(Q_1 - Q_0).$$

Hence

$$Q(t) = (-1, 2, 2) + t(2, 1, -3).$$

Theorem 1.5.1. *The parametric equations of this line are*

$$x(t) = -1 + 2t,$$

$$y(t) = 2 + t,$$

$$z(t) = 2 - 3t.$$

Remark. In general, if a line passes through a point $P_0 = (x_0, y_0, z_0)$ and has direction vector $\mathbf{v} = (a, b, c)$, then a parametric equation for the line is

$$Q(t) = P_0 + tv.$$

Equivalently,

$$x = x_0 + at,$$

$$y = y_0 + bt,$$

$$z = z_0 + ct.$$

The constant terms determine a point on the line, while the coefficients of t determine its direction.

Example 1.5.2 (Cycloid). A circle of radius a rolls along the x -axis without slipping. Find a parametric equation for the path traced by a point on the rim of the circle. Describe the main geometric features of the resulting curve.

Solution. Let t be the angle through which the circle has rotated. Since the circle rolls without slipping, the distance traveled by its center is equal to the arc length at . Therefore the center of the circle is at

$$C(t) = (at, a).$$

Relative to the center, the marked point begins at $(0, -a)$ and rotates clockwise through angle t , so its displacement from the center is

$$(-a \sin t, -a \cos t).$$

Hence the traced curve is

$$Q(t) = (x(t), y(t)) = (a(t - \sin t), a(1 - \cos t)).$$

This curve is called a **cycloid**.

For $0 \leq t \leq 2\pi$, one obtains a single arch of the cycloid. The arch begins at $(0, 0)$, ends at $(2\pi a, 0)$, and reaches its highest point

$$Q(\pi) = (\pi a, 2a).$$

The velocity vector is

$$Q'(t) = (a(1 - \cos t), a \sin t),$$

so for $t \neq 2\pi k$ the slope of the tangent line is

$$\frac{y'(t)}{x'(t)} = \frac{a \sin t}{a(1 - \cos t)} = \sin \frac{t}{1 - \cos t} = \cot\left(\frac{t}{2}\right).$$

Thus the tangent is horizontal at the top point $t = \pi$, while the endpoints $t = 2\pi k$ are cusps.

The arc length of one arch is

$$\int_0^{2\pi} \|Q'(t)\| dt = \int_0^{2\pi} 2a \sin\left(\frac{t}{2}\right) dt = 8a.$$

The area under one arch is

$$\int_0^{2\pi} y(t)x'(t) dt = \int_0^{2\pi} a^2(1 - \cos t)^2 dt = 3\pi a^2.$$

□

2 Differentiation

Here, we examine the concepts of differentiation for functions with more than one variables along with their applications.

2.1 Fundamentals of Partial Derivatives

To visualize a function of two inputs, one may use a 3D graph with z-axis being $f(x, y)$.

When the curve is difficult to be drawn as 3D, contour plot is also a great way for visualization. It allows quick determination the function values, extremum and the growth of the function wrt directions .

Temperature and geographical topological graphs are real life examples of contour plot.

To quantify the change of the growth of the function in x and y directions, we make the following definitions inspired by the one-dimensional differentiation:

$$\frac{\partial f}{\partial x}(x_0, y_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x},$$

and

$$\frac{\partial f}{\partial y}(x_0, y_0) = \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}.$$

In other words, we take the derivative of f wrt x or y while treating the other variable as a constant.

2.2 Geometric Interpretation

For a one variable function, its derivative at a certain point is the slope of the tangent line at that point. In other words, its tangent line at a point could be determined by its derivative at that point.

For a function of two variables, $f(x, y)$, one may similarly define the tangent plane at a certain point $(x_0, y_0, f(x_0, y_0))$. Slice the solid formed by $f(x, y)$ at the point parallel to the x-z plane and y-z plane gives two graphs. With the one variable methods, two tangent lines could be obtained, with slope of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$. Since the tangent plane contains these two lines, the plane's normal could be obtained by crossing the direction of the two lines.

Example 2.2.1. Assume that $f(x, y)$ is a function of two variables with $\frac{\partial}{\partial x}(x_0, y_0) = a$ and $\frac{\partial}{\partial y}(x_0, y_0) = b$. Find the plane tangent to $f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$.

Solution. The line tangent to $f(x_0, y_0)$ that lies in the plane through $(x_0, y_0, f(x_0, y_0))$ parallel to the x-z plane is

$$L_1 = \begin{cases} z = z_0 + a(x - x_0) \\ y = y_0 \end{cases},$$

where $z_0 := f(x_0, y_0)$.

Similarly, the line tangent to $f(x_0, y_0)$ that lies in the plane through $(x_0, y_0, f(x_0, y_0))$ parallel to the y-z plane is

$$L_2 = \begin{cases} z = z_0 + b(y - y_0) \\ x = x_0 \end{cases}.$$

Thus, the tangent plane could be obtained with cross product:

$$z = z_0 + a(x - x_0) + b(y - y_0). \quad \square$$

Theorem 2.2.2. The normal vector wrt a surface that is defined by $\mathbf{r}(u, v)$ is $\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}$, where $\mathbf{r}$ is a 3D vector.

Proof. Notice that $\frac{\partial \mathbf{r}}{\partial v}$ and $\frac{\partial \mathbf{r}}{\partial u}$ are both tangent to surface. Thus, their cross product is normal to the surface. $\square$

2.3 Extremum

One of the most consequential applications of partial derivatives is in optimization problems involving multiple variables, ie. finding the max/min of a multi-variable function.

Theorem 2.3.1. At the local extremum of a differentiable two-variable function $f(x, y)$, $f_x = f_y = 0$. In other words, the tangent plane to f at its local extremum is horizontal.

Note: The global extremum may lie on the boundary or at infinity.

Now, the question is how to determine whether a critical point ($f_x = f_y = 0$) is a maximum or minimum, or neither. In fact, it is possible to have a critical point as a saddle point, that is neither a maximum or minimum.

Thus, a natural question that comes is how to determine what type of extremum a critical point is. For that, we look into the second-derivative test.

2.3.1 Second Derivative Test

Define the second derivatives for a twice-differentiable two-variable function $f(x, y)$, $\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial x^2} = f_{xx}$. We also have $f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = f_{yx}$.

Define the following quantities for the second-derivative test assuming (x_0, y_0) is a critical point,

$$\begin{cases} A = f_{xx}(x_0, y_0) \\ B = f_{xy}(x_0, y_0) \\ C = f_{yy}(x_0, y_0) \end{cases}$$

Then, we have

$$f \text{ has a } \begin{cases} \text{local minimum} & \text{if } AC - B^2 > 0 \wedge A > 0 \\ \text{local maximum} & \text{if } AC - B^2 > 0 \wedge A < 0 \\ \text{saddle point} & \text{if } AC - B^2 < 0 \\ \text{inconclusive} & \text{if } AC - B^2 = 0 \end{cases}.$$

Proof. We invoke the two dimensional Taylor's approximation (which could be obtained by considering the one variable case then use chain rule):

$$\begin{aligned} \Delta f &\approx f_x(x - x_0) + f_y(y - y_0) \\ &\quad + \frac{1}{2}f_{xx}(x - x_0)^2 + f_{xy}(x - x_0)(y - y_0) + \frac{1}{2}f_{yy}(y - y_0)^2. \end{aligned}$$

Since we are at a critical point, $f_x = f_y = 0$. Thus, the determining terms are the second order ones. Therefore,

$$\Delta f \approx \frac{1}{2}f_{xx}(x - x_0)^2 + f_{xy}(x - x_0)(y - y_0) + \frac{1}{2}f_{yy}(y - y_0)^2.$$

Let the perturbation in x be h and the perturbation in y be k . Then,

$$\Delta f \approx \frac{1}{2}f_{xx}h^2 + f_{xy}hk + \frac{1}{2}f_{yy}k^2 := Q(h, k).$$

Observe that

$$Q(h, k) = \frac{1}{2} \begin{pmatrix} h \\ k \end{pmatrix} \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix} \begin{pmatrix} h & k \end{pmatrix}.$$

So, we have the determinant criterion $f_{xx}f_{yy} - f_{xy}^2$. So, if $f_{xx}f_{yy} - f_{xy}^2 > 0$, Q is definite (sign does **not** depend on h and k) – positive when $f_{xx} > 0$ and vice versa. While Q is indefinite (sign **does** depends on h and k) when $f_{xx}f_{yy} - f_{xy}^2 < 0$.

Therefore, for the inconclusive case of $f_{xx}f_{yy} - f_{xy}^2 = 0$, the situation of the critical point depends on higher derivatives. $\square$

Example 2.3.1.1. Consider the function $f(x, y) = x + y + \frac{1}{xy}$, $x \in R^+ \wedge y \in R^+$, find its extremum(s).

Solution. We proceed to find its critical points:

$$\begin{cases} \frac{\partial f}{\partial x} = 1 - \frac{1}{y} \frac{1}{x^2} = 0 \\ \frac{\partial f}{\partial y} = 1 - \frac{1}{x} \frac{1}{y^2} = 0 \end{cases} \Rightarrow \begin{cases} x^2 = y \\ y^2 = x \end{cases} \Rightarrow (x, y) = (1, 1).$$

To determine the status of this critical point, we use the second derivative test.

$$\begin{cases} A = f_{xx} = \frac{2}{y} \frac{1}{x^3} \\ B = f_{xy} = \frac{1}{x^2} \frac{1}{y^2} \\ C = f_{yy} = \frac{2}{x} \frac{1}{y^3} \end{cases}$$

The criterion

$$AC - B^2 = \frac{4}{x^4 y^4} \gg 0 \wedge f_{xx} \gg 0.$$

Thus, $f(x, y)$ has a local minimum at $(1, 1)$ while attains its maximum, ∞ , when $x \rightarrow \infty \vee y \rightarrow \infty \vee (x \rightarrow \infty \wedge y \rightarrow \infty)$. $\square$

2.3.2 Application – Least-Squares Interpolation

Motivation: Given experimental data (x_i, y_i) , what is the “best-fit” line that describes the relationship between x and y .

More rigorously, for n data points of (x_i, y_i) , define the error of a data point to the “best-fit” line, $y = ax + b$ to be $(y(x_i) - y_i)^2$. The best-fit line is the line that minimizes the sum of the errors of the data points. In other words, we aim to minimize

$$D = \sum_{i=1}^n (y_i - (ax_i + b))^2,$$

by manipulating a and b given x_i and y_i .

Naturally, we look for the critical points of D wrt a and b . Differentiating gives

$$\begin{cases} \frac{\partial D}{\partial a} = \sum_{i=1}^n 2(y_i - (ax_i + b))(-x_i) = 0 \\ \frac{\partial D}{\partial b} = \sum_{i=1}^n 2(y_i - (ax_i + b))(-1) = 0 \end{cases}$$

So,

$$\begin{cases} \sum_{i=1}^n x_i y_i = \sum_{i=1}^n x_i (ax_i + b) = \left(\sum_{i=1}^n x_i^2\right)a + \left(\sum_{i=1}^n x_i\right)b \\ \sum_{i=1}^n y_i = \sum_{i=1}^n (ax_i + b) = \left(\sum_{i=1}^n x_i\right)a + nb \end{cases}.$$

Now, we have a linear system of a and b . It remains simple to solve for a and b explicitly.

Moreover, with the second-derivative test, it could be shown that the critical point of a and b obtained above indeed minimizes D rather than maximizes it or is a saddle point.

This method could be used on non-linear fitting as well. For example, for an exponential fitting in the form of $y = ae^{bx}$, where x and y are variables and a and b are constants, a linear form could be obtained by applying $\ln$ on both sides: $\ln(y) = \ln(a) + bx$. From here, we could apply the method of least-squares and solve for a and b .

Additionally, this method could be generalized to any finite number of parameters. With the regression function $f(x, y, z, \dots)$ and

$$D(x, y, z, \dots) = \sum_{i=1}^n (y_i - f(x, y, z, \dots)),$$

a system of linear equations could be obtained by setting $\frac{\partial D}{\partial x} = 0$, $\frac{\partial D}{\partial y} = 0$, etc. Then, it remains to show that the critical value for $(x, y, z, \dots)$ obtained does minimize D .

2.4 Differentials

Motivated by the one dimensional case, if $y = f(x)$, then $dy = f'(x) dx$. From this, an infinitesimal change in one variable is related to another. One obvious benefits of this is that the reverse direction becomes easy to obtain.

Example 2.4.1. If $y = \sin^{-1} x$, find $d\frac{y}{dx}$.

Solution. Applying $\sin$ on both sides, $x = \sin y$, so

$$dx = \cos y dy \implies \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}.$$

□

2.4.1 Total Differentials

Definition 2.4.1.1. For a function $f(x, y, z)$, $x, y, z \in \mathbb{R}$, the total differential of f is

$$df = f_x dx + f_y dy + f_z dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz.$$

However, it is not entirely obvious what are differentials – they are not numbers, matrices, vectors, and they have special rules of manipulations.

1. First, $df \neq \Delta f$. Δf is meant to be a number while df encodes how does change in x , y , and z affect f .
2. Also, df serves as a placeholder for small variables Δx , Δy , and Δz to get $\Delta f \approx f_x \Delta x + f_y \Delta y + f_z \Delta z$.
3. Dividing dt on both sides to find the rate of change wrt t .

$$\frac{df}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} + f_z \frac{dz}{dt}.$$

In summary, $d \neq \Delta$

From rule 3, we invented the chain rule for multi-variables.

2.5 Chain Rule

Firstly, we justify the chain rule. We know that

$$\Delta f \approx f_x \Delta x + f_y \Delta y + f_z \Delta z \implies \frac{\Delta f}{\Delta t} \approx f_x \frac{\Delta x}{\Delta t} + f_y \frac{\Delta y}{\Delta t} + f_z \frac{\Delta z}{\Delta t}.$$

As $\Delta t \rightarrow 0$, $\frac{\Delta f}{\Delta t} \rightarrow \frac{df}{dt}$, $\frac{\Delta x}{\Delta t} \rightarrow \frac{dx}{dt}$, etc. At the limit of $\Delta t \rightarrow 0$, we obtain

$$\frac{df}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} + f_z \frac{dz}{dt}.$$

Example 2.5.1. Given $u(t)$ and $v(t)$. Show that $(uv)' = u'v + uv'$.

Solution. Let $f(u, v) := uv$. Then with chain rule we have

$$\frac{df}{dt} = \frac{d(uv)}{dt} = f_u \frac{du}{dt} + f_v \frac{dv}{dt} = v \frac{du}{dt} + u \frac{dv}{dt}.$$

□

Remark. Quotient rule could also be derived, either as a corollary of product rule with $\frac{u}{v} = u \frac{1}{v}$, or with chain rule for multi-variable.

Polar coordinates poses a motivating case for which x and y are also multi variable functions. Then, chain rule for $f(x, y)$ where $x(u, v)$ and $y(u, v)$ is

$$\begin{aligned} df &= f_x dx + f_y dy \\ &= f_x(x_u du + x_v dv) + f_y(y_u du + y_v dv) \\ &= (f_x x_u + f_y y_u) du + (f_x x_v + f_y y_v) dv \\ &\Rightarrow \begin{cases} \frac{\partial f}{\partial u} = f_x x_u + f_y y_u = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} \\ \frac{\partial f}{\partial v} = f_x x_v + f_y y_v = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v} \end{cases} \end{aligned}$$

Remark. It is tempting to cancel the ∂x and the ∂y . However, these are partial differentials – not total differential nor d . In fact, ∂ should remind us that partial differentials work differently compared to single differentials.

$$\partial \neq d$$

Example 2.5.2. Given

$$\begin{cases} x := r \cos \theta \\ y := r \sin \theta \end{cases}$$

, express

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

in

$$\frac{\partial}{\partial r}, \frac{\partial}{\partial \theta}, \frac{\partial^2}{\partial r^2}, \frac{\partial^2}{\partial \theta^2}.$$

Solution. We have:

$$\frac{\partial \theta}{\partial x} = \frac{\partial \tan^{-1}(\frac{y}{x})}{\partial x} = \frac{-\frac{1}{x^2}}{1 + (\frac{y}{x})^2} = -\frac{1}{x^2 + y^2} = -\frac{1}{r^2},$$

$$\frac{\partial r}{\partial x} = \frac{\partial \sqrt{x^2 + y^2}}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2}} = \frac{x}{\sqrt{x^2 + y^2}} = \frac{x}{r}.$$

So,

$$\begin{aligned} \frac{\partial}{\partial x} &= \frac{\partial}{\partial \theta} \frac{\partial \theta}{\partial x} + \frac{\partial}{\partial r} \frac{\partial r}{\partial x} \\ &= -\frac{y}{r^2} \frac{\partial}{\partial \theta} + \frac{x}{r} \frac{\partial}{\partial r}. \end{aligned}$$

Similarly,

$$\frac{\partial \theta}{\partial y} = \frac{\partial \tan^{-1}(y, x)}{\partial y} = \frac{x}{r^2},$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}.$$

Thus,

$$\begin{aligned}\frac{\partial}{\partial y} &= \frac{\partial \theta}{\partial y} \frac{\partial}{\partial \theta} + \frac{\partial r}{\partial y} \frac{\partial}{\partial r} \\ &= \frac{x}{r^2} \frac{\partial}{\partial \theta} + \frac{y}{r} \frac{\partial}{\partial r}.\end{aligned}$$

$$\begin{aligned}\nabla^2 &= \frac{\partial}{\partial x} \frac{\partial}{\partial x} + \frac{\partial}{\partial y} \frac{\partial}{\partial y} \\ &= \left(-\frac{y}{r^2} \frac{\partial}{\partial \theta} + \frac{x}{r} \frac{\partial}{\partial r} \right)^2 + \left(\frac{x}{r^2} \frac{\partial}{\partial \theta} + \frac{y}{r} \frac{\partial}{\partial r} \right)^2 \\ &= \left(-\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} + \cos \theta \frac{\partial}{\partial r} \right)^2 + \left(\frac{\cos \theta}{r} \frac{\partial}{\partial \theta} + \sin \theta \frac{\partial}{\partial r} \right)^2 \\ &= -\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(-\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} + \cos \theta \frac{\partial}{\partial r} \right) \\ &\quad + \cos \theta \frac{\partial}{\partial r} \left(-\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} + \cos \theta \frac{\partial}{\partial r} \right) \\ &\quad + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \left(\frac{\cos \theta}{r} \frac{\partial}{\partial \theta} + \sin \theta \frac{\partial}{\partial r} \right) \\ &\quad + \sin \theta \frac{\partial}{\partial r} \left(\frac{\cos \theta}{r} \frac{\partial}{\partial \theta} + \sin \theta \frac{\partial}{\partial r} \right) \\ &= -\frac{\sin \theta}{r} \left(-\frac{\cos \theta}{r} \frac{\partial}{\partial \theta} - \frac{\sin \theta}{r} \frac{\partial^2}{\partial \theta^2} - \sin \theta \frac{\partial}{\partial r} + \cos \theta \frac{\partial^2}{\partial r \partial \theta} \right) \\ &\quad + \cos \theta \left(\frac{\sin \theta}{r^2} \frac{\partial}{\partial \theta} - \frac{\sin \theta}{r} \frac{\partial^2}{\partial r \partial \theta} + \cos \theta \frac{\partial^2}{\partial r^2} \right) \\ &\quad + \frac{\cos \theta}{r} \left(-\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} + \frac{\cos \theta}{r} \frac{\partial^2}{\partial \theta^2} + \cos \theta \frac{\partial}{\partial r} + \sin \theta \frac{\partial^2}{\partial r \partial \theta} \right) \\ &\quad + \sin \theta \left(-\frac{\cos \theta}{r^2} \frac{\partial}{\partial \theta} + \frac{\cos \theta}{r} \frac{\partial^2}{\partial r \partial \theta} + \sin \theta \frac{\partial^2}{\partial r^2} \right).\end{aligned}$$

Combining like terms gives,

$$\begin{aligned}
&= \left(\frac{\sin^2 \theta}{r^2} + \frac{\cos^2 \theta}{r^2} \right) \frac{\partial^2}{\partial \theta^2} + (\cos^2 \theta + \sin^2 \theta) \frac{\partial^2}{\partial r^2} \\
&\quad + \left(-\frac{\sin \theta \cos \theta}{r} - \frac{\sin \theta \cos \theta}{r} + \frac{\sin \theta \cos \theta}{r} + \frac{\sin \theta \cos \theta}{r} \right) \frac{\partial^2}{\partial \theta \partial r} \\
&\quad + \left(\frac{\sin^2 \theta}{r} + \frac{\cos^2 \theta}{r} \right) \frac{\partial}{\partial r} \\
&\quad + \left(\frac{\sin \theta \cos \theta}{r^2} + \frac{\sin \theta \cos \theta}{r^2} - \frac{\sin \theta \cos \theta}{r^2} - \frac{\sin \theta \cos \theta}{r^2} \right) \frac{\partial}{\partial \theta} \\
&= \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r}.
\end{aligned}$$

□

Remark. Notice in here, we abused the notation as if $\frac{\partial^2 a}{\partial b^2} = \frac{\partial}{\partial b} \frac{\partial}{\partial b} a = \left(\frac{\partial}{\partial b} \right)^2 a$. However, if this abuse is taken further, treating $\frac{\partial^2 a}{\partial b^2}$ as $\frac{\partial a}{\partial b} \cdot \frac{\partial a}{\partial b}$, it is not necessarily true, especially with the constants. If $\frac{\partial^2 a}{\partial b^2}$ is *directly* treated as multiplication, then the third equality will follow as:

$$\begin{aligned}
&= \frac{y^2}{r^4} \frac{\partial^2}{\partial \theta^2} + \frac{x^2}{r^2} \frac{\partial^2}{\partial r^2} - \frac{2xy}{r^3} \frac{\partial \theta}{\partial r} \\
&\quad + \frac{x^2}{r^4} \frac{\partial^2}{\partial \theta^2} + \frac{y^2}{r^2} \frac{\partial^2}{\partial r^2} + \frac{2xy}{r^3} \frac{\partial^2}{\partial \theta \partial r} \\
&= \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial r^2},
\end{aligned}$$

which misses the $\frac{1}{r} \frac{\partial}{\partial r}$ term.

2.6 Gradient

Definition 2.6.1. Fix a three variable real function $f(x, y, z) = w$, define

$$\nabla f = (f_x, f_y, f_z).$$

Observe that now we could write

$$\frac{df}{dt} = \nabla f \cdot \frac{d\mathbf{r}}{dt}.$$

Theorem 2.6.2. Let $w = f(x, y, z)$, then $\nabla w(x_0, y_0, z_0) \perp$ the level surface $\mathcal{S}$ at (x_0, y_0, z_0) , described by $f(x, y, z) = f(x_0, y_0, z_0)$, denoted as $\mathcal{S}$.

Proof. Consider a curve described by $\mathbf{r} = \mathbf{r}(t) \in \mathcal{S}$. By chain rule,

$$\frac{dw}{dt} = \nabla w \cdot \frac{d\mathbf{r}}{dt} = 0.$$

Thus, $\nabla w \perp \frac{d\mathbf{r}}{dt}$. However, $\frac{d\mathbf{r}}{dt}$ is tangent to the curve therefore tangent to $\mathcal{S}$. Since $\mathbf{r} = \mathbf{r}(t)$ is arbitrary, ∇w is tangent to all tangent vectors of $\mathcal{S}$ at (x_0, y_0, z_0) , thus tangent to $\mathcal{S}$ at (x_0, y_0, z_0) . $\square$

Example 2.6.3. Find the tangent plane to $x^2 + y^2 - z^2 = 4$ at $(2, 1, 1)$.

Solution. The normal to the plane is ∇w where $w = f(x, y, z)$.

$$\nabla w = (f_x, f_y, f_z) = (2x, 2y, -2z) = (4, 2, -2).$$

Thus, the plane is

$$4x + 2y - 2z = 4 \cdot 2 + 2 \cdot 1 - 2 \cdot 1 = 8.$$

$\square$

We continue our discussion with gradient with directional derivatives.

2.7 Directional Derivative

In gradient and partial derivatives, we considered the change of the function along the direction of x, y, z , etc. Now, we consider the change in the function along a customized, non-coordinate direction.

In other words, fix a real function $z = w(x, y)$, and a two dimensional vector in the x-y plane $\hat{\mathbf{u}}$, which is parameterized by s . $\frac{d\mathbf{r}}{ds} = \hat{\mathbf{u}}$, what is $\frac{dw}{ds}$

Definition 2.7.1. The directional derivative of the function w wrt arc-length s in the direction of $\hat{\mathbf{u}}$ is

$$\left. \frac{dw}{ds} \right|_{\hat{\mathbf{u}}}.$$

In other words, $\left. \frac{dw}{ds} \right|_{\hat{\mathbf{u}}}$ is the slope of the slice of graph by a vertical plane parallel to $\hat{\mathbf{u}}$.

With chain rule,

$$\left. \frac{dw}{ds} \right|_{\hat{\mathbf{u}}} = \nabla w \cdot \hat{\mathbf{u}}.$$

As a quick sanity check,

$$\left. \frac{dw}{ds} \right|_{\hat{\mathbf{i}}} = \nabla w \cdot \hat{\mathbf{i}} = \frac{\partial w}{\partial x}.$$

Furthermore,

$$\nabla w \cdot \hat{u} = |\nabla w| \cos \theta.$$

Since $\cos \theta \leq 1$, this implies the increment of the function is maximized along the direction of the gradient, minimized in the opposite direction as the gradient. Also, $\hat{u} \perp \nabla w \iff \hat{u}$ is tangent to level.

2.8 Lagrange Multipliers

Constraint optimization is a core theme in many sciences and maths. When dealing with continuous changes, differentiation is one of the most important ways of dealing with them; however, our current method of critical points and the second derivative test is not enough to model *constraints*. Thus, we present lagrange multipliers.

Formally, lagrange multipliers method is a way of finding the extrema of a smooth, multi-variate function $f(x, y, z)$ where x, y, z are dependent described by the constraint equation $g(x, y, z) = c$, where c is a constant.

The key observation here is that at extremum, the level curve of f is tangent to g . In other words, their normal vectors are parallel. Expressed in terms of gradient,

$$\nabla f = \lambda \nabla g$$

3 Integration

Here, we examine the concepts of integration for functions with more than one variables along with their applications.

A Appendix

A.1 Kepler's Laws and Elliptic Orbits

Planetary motion gives a natural application of parametrized and polar curves. Kepler's laws say that planets move in ellipses, that the radius vector from the Sun sweeps out equal areas in equal times, and that the orbital period is tied to the size of the orbit.

Definition 1.1.1. Kepler's first law. A planet moves in an ellipse with the Sun at one focus.

Definition 1.1.2. Kepler's second law. If $A(t)$ denotes the area swept out by the radius vector from the Sun to the planet, then

$$A'(t)$$

is constant.

Definition 1.1.3. Kepler's third law. For planets orbiting the same Sun,

$$T^2 = ka^3,$$

where T is the orbital period, a is the semimajor axis of the orbit, and k is the same constant for every planet orbiting that Sun.

If an ellipse has semimajor axis a and eccentricity e , and if polar coordinates are measured from a focus, then its equation can be written as

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta}.$$

This is especially useful in celestial mechanics because the Sun is located at a focus rather than at the center of the ellipse.

The second law is equivalent to the statement that the **areal velocity** is constant:

$$A'(t) = \frac{1}{2}r^2\theta'(t).$$

Therefore a planet moves faster when it is closer to the Sun and slower when it is farther away.

Remark. A centered parametrization of the ellipse is

$$x(t) = a \cos t, \quad y(t) = b \sin t.$$

To model a planetary orbit with the Sun at a focus, one shifts the origin from the center of the ellipse to the focus and then uses the focus-based polar equation above.

B Surfaces

B.1 Quadratic Surfaces

The most general **quadratic surface** (also called a **quadric surface**) in $\mathbb{R}^3$ is given by

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0,$$

where $A, B, C, D, E, F, G, H, I, J$ are constants, not all of A, B, C, D, E, F equal to zero.

This can be written in matrix form as

$$\begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} A & \frac{D}{2} & \frac{E}{2} \\ \frac{D}{2} & B & \frac{F}{2} \\ \frac{E}{2} & \frac{F}{2} & C \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} G & H & I \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + J = 0.$$

The mixed terms Dxy , Exz , and Fyz correspond to rotations of the coordinate axes. By translating and rotating coordinates, every nondegenerate quadric can be reduced to one of the following standard forms.

B.2 Ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

B.3 Hyperboloid of One Sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

B.4 Hyperboloid of Two Sheets

$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

B.5 Elliptic Cone

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$

B.6 Elliptic Paraboloid

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

B.7 Hyperbolic Paraboloid

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

B.8 Elliptic Cylinder

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

B.9 Hyperbolic Cylinder

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

B.10 Parabolic Cylinder

$$y = \frac{x^2}{a^2}$$

Remark. Every quadratic surface is the zero set of a polynomial of degree two in the variables x , y , and z .